



COURSE MATERIAL

Interactive Diagram Demo: Recursion

Build and unwind the factorial call stack

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Reading the sequence

The growing activation bars represent pending stack frames; the dashed arrows show values returning as the stack unwinds.

The submission buttons above are placeholders. They open `example.com` and do not submit data.

Evaluating factorial(3)

Each call waits for the next smaller factorial before it can return.

Call factorial(3)

Main starts the recursive calculation.

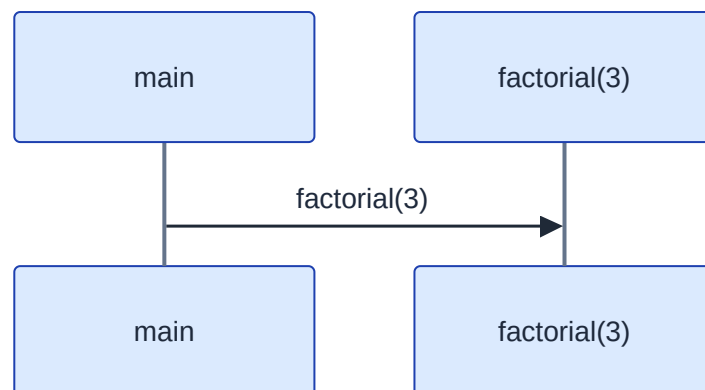


Figure 1: Call factorial(3)

Recurse to factorial(2)

factorial(3) needs the value of factorial(2).

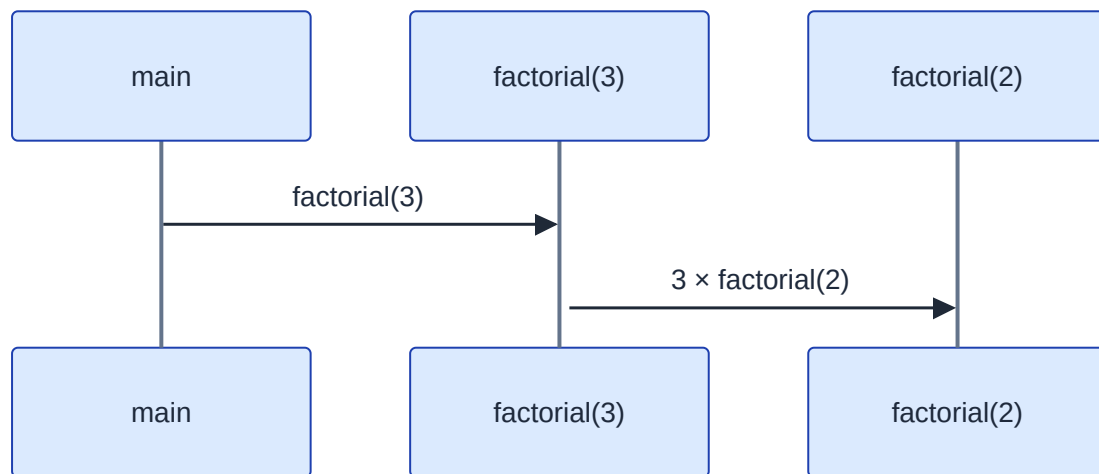


Figure 2: Recurse to factorial(2)

Reach the base case

factorial(2) calls factorial(1), which returns 1.

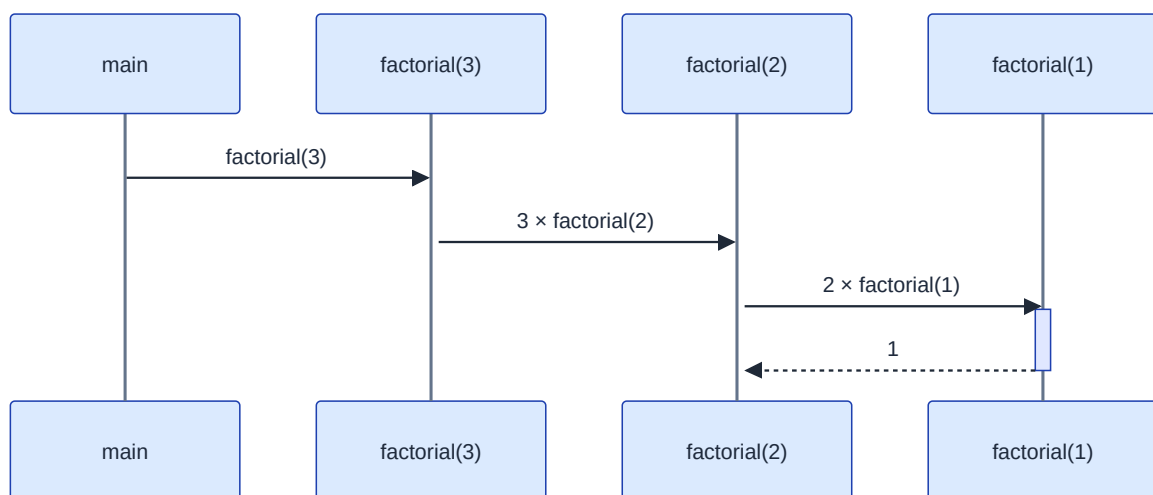


Figure 3: Reach the base case

Unwind factorial(2)

factorial(2) multiplies 2 by 1 and returns 2.

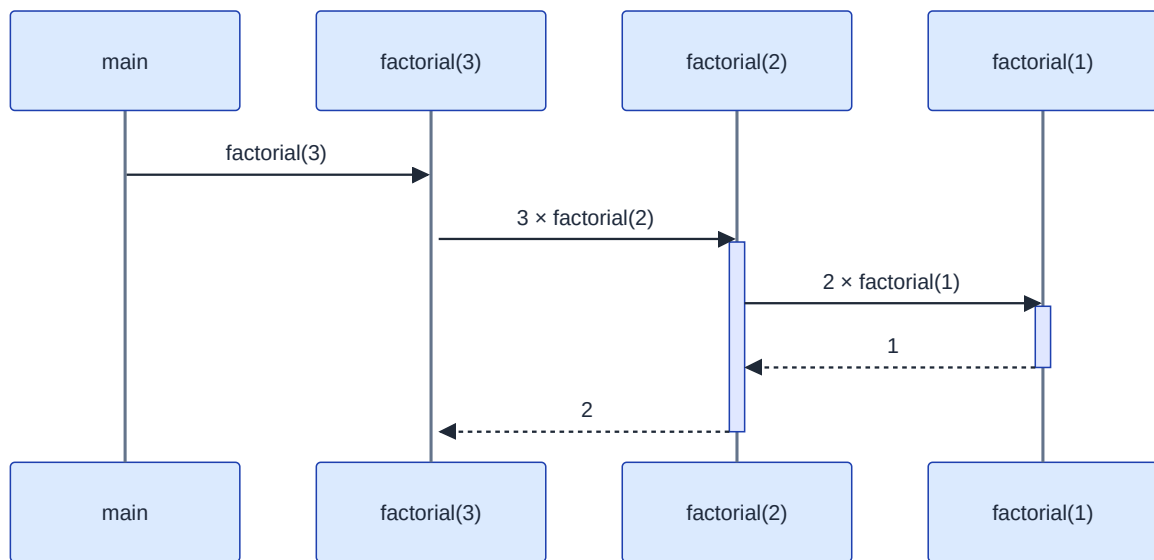


Figure 4: Unwind factorial(2)

Return the final result

`factorial(3)` multiplies 3 by 2 and returns 6 to `main`.

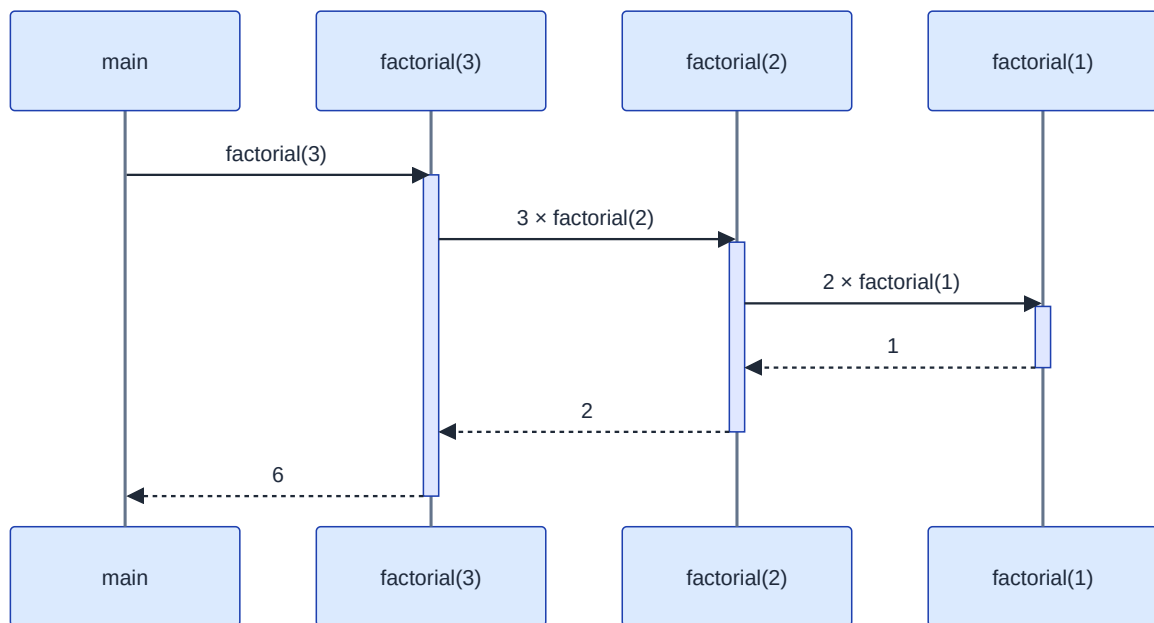


Figure 5: Return the final result